Tong
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REVIEW

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A Narrative Review of Effects and Underlying Mechanisms of Needling Therapy for Treating Myofascial Pain and Musculoskeletal Disorders

Cheuk Ming Tong[1*]

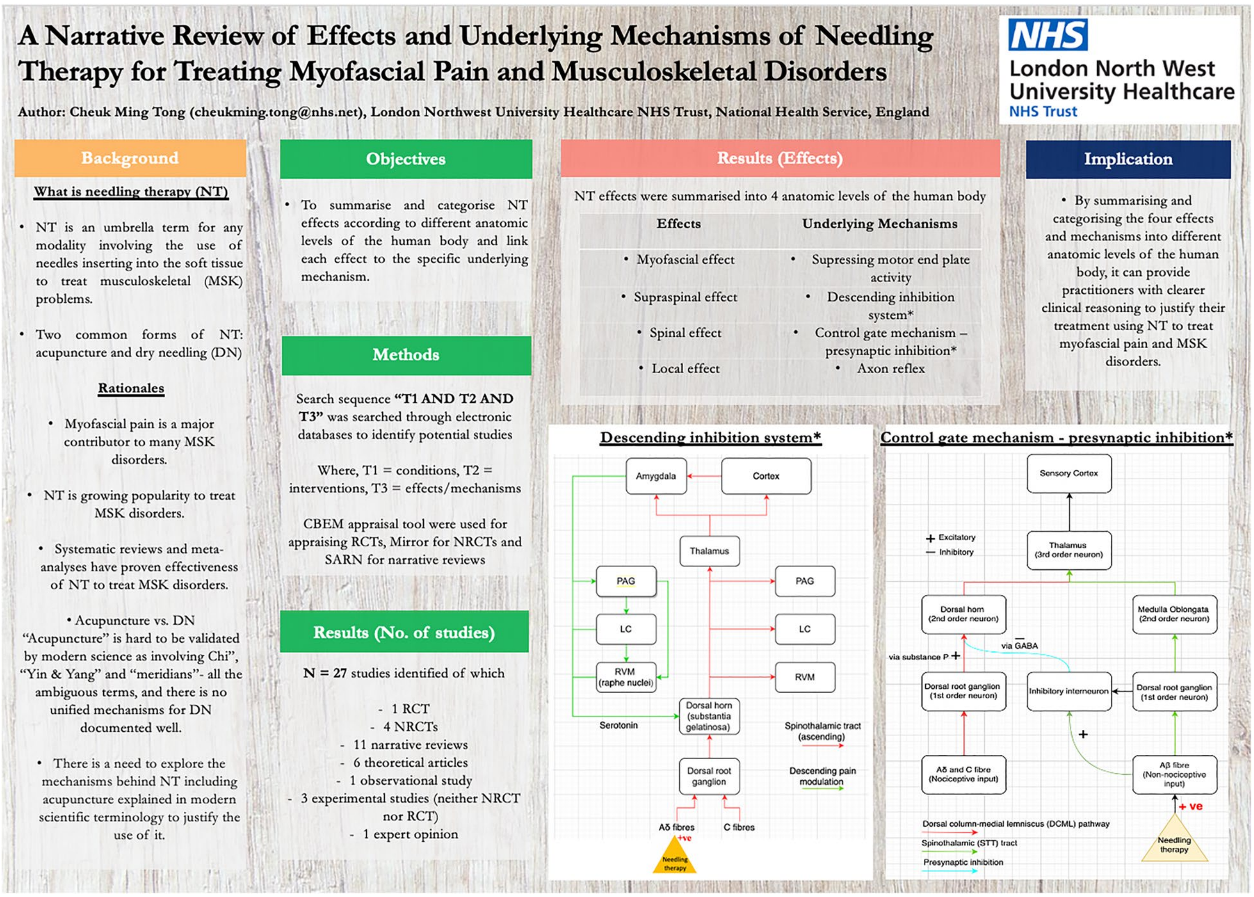
Abstract

This review aims to summarise the effects and underlying mechanisms of needling therapy (NT) in the treatment of myofascial pain and musculoskeletal (MSK) disorders. This review aims doing so by categorising NT effects according to different anatomic levels of the human body and linking each effect to the specific mechanism. Acupuncture, the first form of NT, has existed for 3000 years and been practised by Traditional Chinese Medicine (TCA); however, its mechanism is hard to be validated by modern science as it involves "Chi", "Yin & Yang" and "meridians"—all the ambiguous concepts which are hard to be interpreted by the West and modern science. Dry needling (DN), another form of NT, which emerged in the West in the twentieth century, has not well been explicitly reviewed for its mechanisms. Given that NT—acupuncture originated in the East and DN originated in the West—has been proven by systematic reviews and meta-analyses for its effectiveness to treat MSK disorders (world's major disability contributor as defined by World Health Organization) and that many MSK disorders are caused by myofascial pain, a summary of NT effects and mechanisms in modern scientific terminology can provide practitioners with clearer clinical reasoning to justify their treatment using NT to treat myofascial pain and MSK disorders. This review summarised 4 main effects according to different anatomic levels of the human body attributed to NT. They are myofascial effect, supraspinal effect, spinal effect and local effect. The mechanisms responsible for each of these effects, respectively, are supressing motor end plate activity, descending inhibition mechanism, control gate mechanism and axon reflex.

Keywords Needling therapy, Acupuncture, Dry needling, Myofascial pain, Musculoskeletal disorders, Mechanisms, Effects

*Correspondence:
Cheuk Ming Tong
cheukming.tong@nhs.net

Graphical Abstract



1 Introduction

Musculoskeletal (MSK) disorders are defined as conditions affecting muscles, ligaments, tendons, bones and nerves acutely and chronically [1]. Data from the Global Burden of Disease shows that MSK disorders are the most prevalent category of health issue affecting nearly 1.71 billion people worldwide [2]. In the United Kingdom, almost one third of the population suffer from MSK disorders [3]. The World Health Organization has defined MSK disorders as the leading contributor to disability worldwide [4]. While the most prevalent MSK disorders is low back pain [4], 85% of low back pain is caused by myofascial pain [5], and myofascial pain accounts for 30–85% of other different MSK disorders [6]. A systematic review conducted by Babatunde et al. [7] has shown that exercise therapy, psychosocial interventions and pharmacological treatments are effective to treat MSK disorders. However, these commonly practised modalities appear to be insufficient to lower the high prevalence of MSK disorders.

Needling therapy (NT), therefore, starts to gain its popularity as a new modality to treat MSK disorders due to growing evidence to prove its efficacy. NT is an umbrella term for any modality involving the use of needles inserting into the soft tissue to treat MSK pain and conditions [8]. There are two common forms of NT: acupuncture and dry needling (DT). While DT is defined by the American Physical Therapy Association (APTA) as a skilled intervention involving the use of a filiform needle to penetrate the skin to stimulate underlying myofascial points, muscular and connective tissues to treat MSK pain and movement impairments [9], acupuncture (translated as needle penetration) is defined in TCM as the use of small needles inserted into the specific points of the body (acupoints points) to treat bodily problems [10]. Two systematic reviews have shown effectiveness of NT to treat low back pain (LBP) [11, 12]. One systematic review and meta-analysis has shown effectiveness of NT combined with stretching to treat myofascial trigger points (TrP) [13]. Other systematic reviews and meta-analyses have also shown effectiveness of NT to treat temporomandibular disorder [14], frozen shoulder [15], stiff neck [16] and plantar fasciitis [17]. These modern evidences proven in the West demonstrate the effectiveness of NT to treat MSK disorders, and the first form of NT—acupuncture—actually has been practised in the East for almost 3000 years [18].

The first documentation of acupuncture was in "Nuangdi Neijing", which is translated as "the Inner Canon of the Yellow Emperor". It was a book written in about 2000 years ago in the Han Dynasty and is the fundamental doctrine for traditional Chinese Medicine (TCM) [19]. This book has documented the mechanism of acupuncture to explain how it works to treat all bodily pains and conditions. The underlying mechanism of acupuncture, according to the book, is to balance the "Yin" and "Yang" of a human body to restore internal harmony. "Yin" and "Yang" are both important opposing concepts in TCM. While "Yin" symbolises female, cold, quiet, etc., "Yang" symbolises male, warm, movement, etc. The human body has 20 meridian channels, where contain "Chi" to flow. "Chi", another important concept in TCM, can be translated as "vital energy". While "Chi" flows through the meridian channels, and "blood" flows through the blood vessels. Blood contributes to "Yin", and "Chi" contributes to "Yang". Acupuncture has an effect on "Yang". Since "Yin" and "Yang" has to be balanced well in the human body to achieve internal harmony, all bodily pains and conditions occur once human body loses internal harmony (i.e., "Yin" and "Yang" are not balanced), and acupuncture works by modulating "Yang" in the human body to achieve internal harmony. Once internal harmony is achieved, all diseases are gone (Fig. 1).

However, the mechanism of acupuncture explained by TCM all depends on "Chi". Without the proof of the existence of "Chi", the whole theory does not stand. Unfortunately, 2000 years later, "Chi" still has not been proven yet. This theory, therefore, is not likely to be accepted in the current literature. As a result, the mechanisms behind NT including acupuncture explained in modern scientific terminology need to be explored to justify the use of it. In addition, NT has grown in popularity over the last 20 years to treat myofascial pain and MSK disorders [20]; however, no unified theories and mechanisms have well been reviewed and documented in the current literature—most of them are rather equivocal. This review is the first study aiming to summarise and categorise NT effects according to different anatomic levels of the human body and to link each effect to the specific underlying mechanism to treat myofascial pain and MSK disorders. The terms "effect" and "mechanism" are causally related, so by unifying them, it can provide practitioners with clearer clinical reasoning to justify their treatment using NT to treat myofascial pain and MSK disorders.

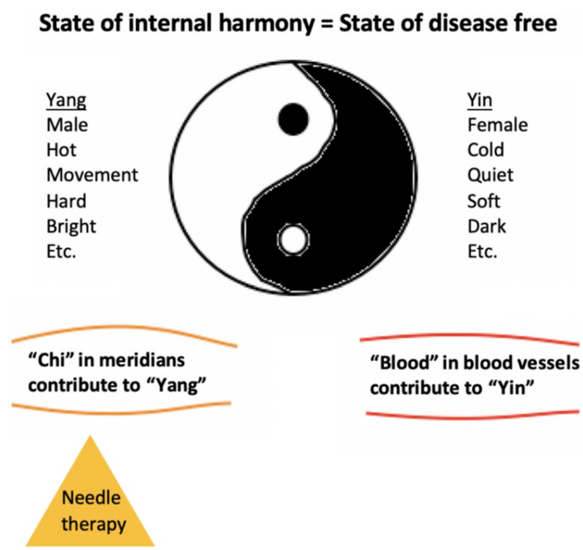


Fig. 1 State of internal harmony with concept of Yin and Yang according to TCM

2 Methods

The methodology of narrative review was adopted for the conduction of this review as the scope of this study is of a border topic with an intention of synthesising and summarising the current literature on effects and mechanisms of NT (acupuncture and dry needling) for treating wider population of conditions which are myofascial pain and MSK disorders.

Four electronic databases (AMED, CLINAL, Cochrane Library and Medline) were searched to identify the potential studies. External potential studies were also identified through manual searching of citation from the included studies. Three main themes comprising the search strategy were defined (conditions, interventions and effects/mechanisms). Synonyms and words of similar concepts in each theme were connected using Boolean operator "OR". The final search sequence was then created by connecting each theme with Boolean operators ("AND"). Detailed search strategy was listed in appendix 1. Exclusion criteria were duplicate studies, studies without full text, not in English and conducted before 2005 and studies with non-available references. In addition, this review is only interested in NT that does not involve injection of substance; therefore, any studies focus on wet needling were also excluded. The search years were from 2005 to 2025 (manual citation search not limited to years). Studies were not limited to RCTs, systematic